Energy Research & Social Science 108 (2024) 103405

Contents lists available at ScienceDirect

Energy Research & Social Science

journal homepage: www.elsevier.com/locate/erss

Review

Youth as energy citizens or passive actors? A critical review of energy transition scholarship

Arwa Jaradat [a], Bram Noble [b,*], Greg Poelzer [c]

[a] Department of Geography and Planning, 117 Science Place, University of Saskatchewan, Saskatoon, Saskatchewan S7N 5C8, Canada
[b] Department of Geography and Planning, University of Saskatchewan, Saskatoon, Saskatchewan, Canada
[c] School of Environment and Sustainability, University of Saskatchewan, Saskatoon, Saskatchewan, Canada

ARTICLE INFO

Keywords:
Youth
Energy transition
Energy citizens
Engagement
Renewable energy

ABSTRACT

Youth are increasingly recognized in global arenas as critical actors for achieving sustainable development goals, including just energy transitions. Many scholars similarly argue the importance of engaging youth in energy transitions; yet, there is limited understanding of whether and how youth are framed as legitimate actors in energy transitions scholarship. This paper examines how recent energy transitions research conceptualizes youth, specifically whether youth are approached as influential energy citizens or merely as passive actors. Results show that the majority of research does not frame youth as energy citizens per se; rather, the dominant focus is on energy literacy and the importance of educating youth about energy options. This is followed by scholarship exploring youth perspectives on, or testing youth knowledge about, different energy technologies, youth energy conservation behaviour, and youth climate advocacy. Limited scholarship has focused on the role and influence of youth in energy governance. Research is needed at the national to international scale, focused on the rationale, value added and influence of youth in the multi-lateral energy space, and at local or community scale, enabling youth's roles in local energy planning and renewable energy projects.

1. Introduction

Transitioning from centralized, fossil fuel-based energy systems toward renewable and decentralized energy is recognized globally as essential to mitigating climate change and providing affordable, reliable, and accessible energy. Energy systems are often framed as socio-technical systems, whereby technology is a social phenomenon embedded in infrastructures, shared by human actors, and used among social networks [1]. Transitions in energy systems involve multi-dimensional shifts from one mode of production and consumption to another [2] and are accompanied by changes in technologies, institutional arrangements, and social networks as new actors and new roles are created [1,3,4]. The success of energy transitions thus hinges in large part on the engagement and support of a diversity of energy actors, from producers and consumers to innovators and policymakers, over an extended period of time [2,5].

Energy citizenship can be a critical driver and sustainer of energy transitions [6,7]. Energy citizenship emphasizes individuals as agents of change by acting responsibly toward energy consumption [8], being aware of the types of energy they are using [9], engaging in energy policy and decision-making arenas through democratic and civil opportunities [10,11], and participating in energy and climate advocacy [7]. Various scholars have emphasized the importance of energy citizenship for an inclusive, democratic, and just energy transition and for reinforcing the legitimacy and acceptance of new energy technologies or agendas [7,12,13]. Energy citizens are also characterized as the front-runners of innovation [14,15], providing important transition arenas that can ultimately influence energy regimes [16,17], and foundational to transformative change. Yet, notwithstanding a growing body of research on energy citizenship addressing such topics as gender [18], local energy cooperatives [19], communities of place [20], and engagement of historically poor and marginalized populations [21], Silvast and Valkenburg [22] argue that the socio-economic positioning of energy citizens is poorly understood and there remains a need for critical discussion as to "who counts as a citizen."

This paper is based on the premise that youth are essential and legitimate citizens in the process of a just and inclusive energy transition [23,24], and a critical catalyst for the deployment of local community renewable energy [25]. There is no single definition of what constitutes "youth", but youth are often characterized as individuals between 15

* Corresponding author.
E-mail addresses: atj779@usask.ca (A. Jaradat), b.noble@usask.ca (B. Noble), greg.poelzer@usask.ca (G. Poelzer).

https://doi.org/10.1016/j.erss.2023.103405
Received 6 September 2023; Received in revised form 23 November 2023; Accepted 19 December 2023
Available online 27 December 2023
2214-6296/© 2023 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

and 30 years of age [26–28]. The United Nations' [29] world youth report identifies youth as central to finding solutions for complex sustainability challenges, including implementation of all 17 Sustainable Development Goals (SDGs). The International Renewable Energy Agency (IRENA) similarly emphasizes youth as a pivotal part of the global energy transition and has hosted multiple forums where youth leaders gather to discuss energy transitions, including the promotion of inclusive and just energy transitions [24]. However, notwithstanding advances among the international policy community on youth engagement, coupled with literature arguing for the need to engage youth in energy and climate action, there remains limited understanding of how youth are framed in current energy transitions scholarship. Scholars have discussed the importance of youth in the implementation of SDGs [30,31], especially education and employment [32,33] and also climate action [34–36], but we are not aware of any research that specifically explores whether and how youth are framed as energy citizens in global energy transition.

This paper responds to this gap in energy scholarship and examines how the literature conceptualizes the role of youth in energy transitions. Specifically, we are interested in whether energy scholarship approaches youth as influential energy citizens and drivers of energy innovation, or whether youth are framed only as passive actors. In the sections that follow our approach for examining the energy literature is presented first, followed by a thematic analysis of energy research that focuses on or engages youth. Several observations are then presented concerning the state of scholarship, followed by recommendations for research on youth energy citizenship in energy transitions.

2. The role of youth in the energy sector

Scholarship addressing youth in the energy sector is not new or unique to renewable energy. In the 1990s, for example, Seydlitz et al. [130] explored the implications of the offshore oil industry on community wellness, including youth, while Seyfrit and Hamilton [131] examined young people's out-migration intentions from North Sea oil-dependent Scottish island communities. More recent scholarship has explored youth identities in oil-dependent communities in Canada [132], the expectations among youth in developing countries for new-found employment associated with the prospects of oil development [133], and even the marginalization of youth from participating in the workforce in emerging fossil fuel economies [134]. The emphasis of such scholarship is often on the disproportionate vulnerability of, and adverse impacts on youth in fossil fuel-dependent regions [135], coupled with opportunities for youth engagement in energy sector employment [136], rather than the role and contribution of youth to energy discourse and energy futures per se [137].

Energy systems globally are transitioning, from centralized and vertically integrated systems reliant on oil, natural gas, and coal, toward more distributed and local energy systems powered by renewables [20,126]. Associated with this transition is a new role for civil society, including youth, in shaping an inclusive and just energy transition and ensuring socially acceptable and beneficial energy futures [7,138,139]. In the context of rural and remote communities, for example, McMaster et al. [25,56] argue that youth must be approached as critical drivers of energy innovation, as their engagement is foundational to the long-term success of any socio-technical transition. Although youth have often been excluded from energy decision making [137,139], they are now claiming "their power as mobilized citizens" to influence both energy and climate agendas and to challenge the status quo of power relations [139]. Yet, it is only in recent years that government agencies and energy organizations have been seeking to actively involve youth in their activities [71].

Sovacool et al. [140] argue the need for greater attention to understanding the acceleration of transitions and their embeddedness in society. This may require a reconceptualization of youth's roles in the energy sector [55], establishing the need for a more critical account of how scholars approach youth in energy transition research [137] – whether as mobilized citizens or as passive actors.

3. Methods

A systematic literature analysis was conducted based on the Scopus database. Scopus is a scholarly database organized by a distinctive indexing system containing peer-reviewed literature, trade publications, conference proceedings, patent records and Websites, from 1996 onward [37,38]. Scopus was selected due to its coverage, indexing system, and the ability to replicate research queries [39–43]. Determining a comprehensive search string involved an iterative process, due to the variety of terms with similar meaning as *youth* and the diversity of terminology related to renewable energy. Working with a library information scientist, the following search string was adopted:

[TITLE-ABS-KEY (youth OR adolescent OR teenage OR youngster OR "young people" OR "young citizens" AND energy AND "renewable energy" OR solar OR wind OR biofuel OR biomass OR bioenergy OR geothermal OR nuclear OR electricity OR "clean energy" OR "alternative energy" OR "green energy" OR "sustainable energy" OR transition OR future OR scenario].

This produced 4997 results of different document types in disciplines ranging from medicine to applied engineering (Fig. 1). The search was then limited to English-language peer-reviewed journal articles in Scopus subject areas of social sciences, energy, environmental sciences, and engineering, as we presumed these areas would have the most plausible connections to the topics of youth and renewable energy. This narrowing of the search parameters resulted in 531 articles.

Results were then limited to articles published between 2005 and 2022. Narrowing the temporal scope to articles published 2005 onward was to capture research in the decade prior to adoption of the 2030 Agenda for Sustainable Development, adopted by UN member states in 2015, and research that has emerged in the near decade since. The UN World Youth Report under the 2030 Agenda [29] emphasizes international and national actions to ensure youth engagement and participation in implementing global sustainable development goals (SDGs), including sustainable energy. This filtering generated 457 articles. There is a limited amount of energy scholarship published prior to 2005 that addresses youth, often focused on youth training or employment [141,142], and indeed scholarship that exists prior to the temporal scope of coverage of the Scopus database (i.e., prior to 1996) – usually focused on the impact of energy development (namely oil and gas) on youth, such as social health or out-migration [130,131].

Titles and abstracts were then scanned for all 457 articles to exclude any studies that had limited to no direct relevance to the research topic, such as articles focused on youth health and energy (i.e., metabolism, nutrition) and technical engineering or design topics, yielding 127 articles. The remaining articles were then reviewed to eliminate studies that focused on public acceptance of renewable energy but with no concentration on youth, or those focused on household choices about energy use but without specifying youth choices. Papers containing the term *youth* without studying youth per se, such as the energy efficiency of a *youth center* or building, were also excluded leaving a total of 91 papers for detailed analysis. A summary of the screening process is shown in Fig. 1.

Articles were then reviewed in their entirety and coded thematically using NVivo v.12 qualitative analysis software. An iterative coding process was adopted, following the overall approach used by Leonhardt et al. [44] to explore government instruments for energy transition, and based on guidance for thematic analysis in qualitative research [45] and applications of thematic coding in other research fields [46]. In the analysis, topics are defined as the specific areas or foci of research and themes as connecting multiple topics. Papers were first reviewed and coded to identify the key topics addressed in each paper, such as youth energy literacy, behaviour, awareness, perceptions, and attitudes regarding specific energy technologies or programs. Articles were coded

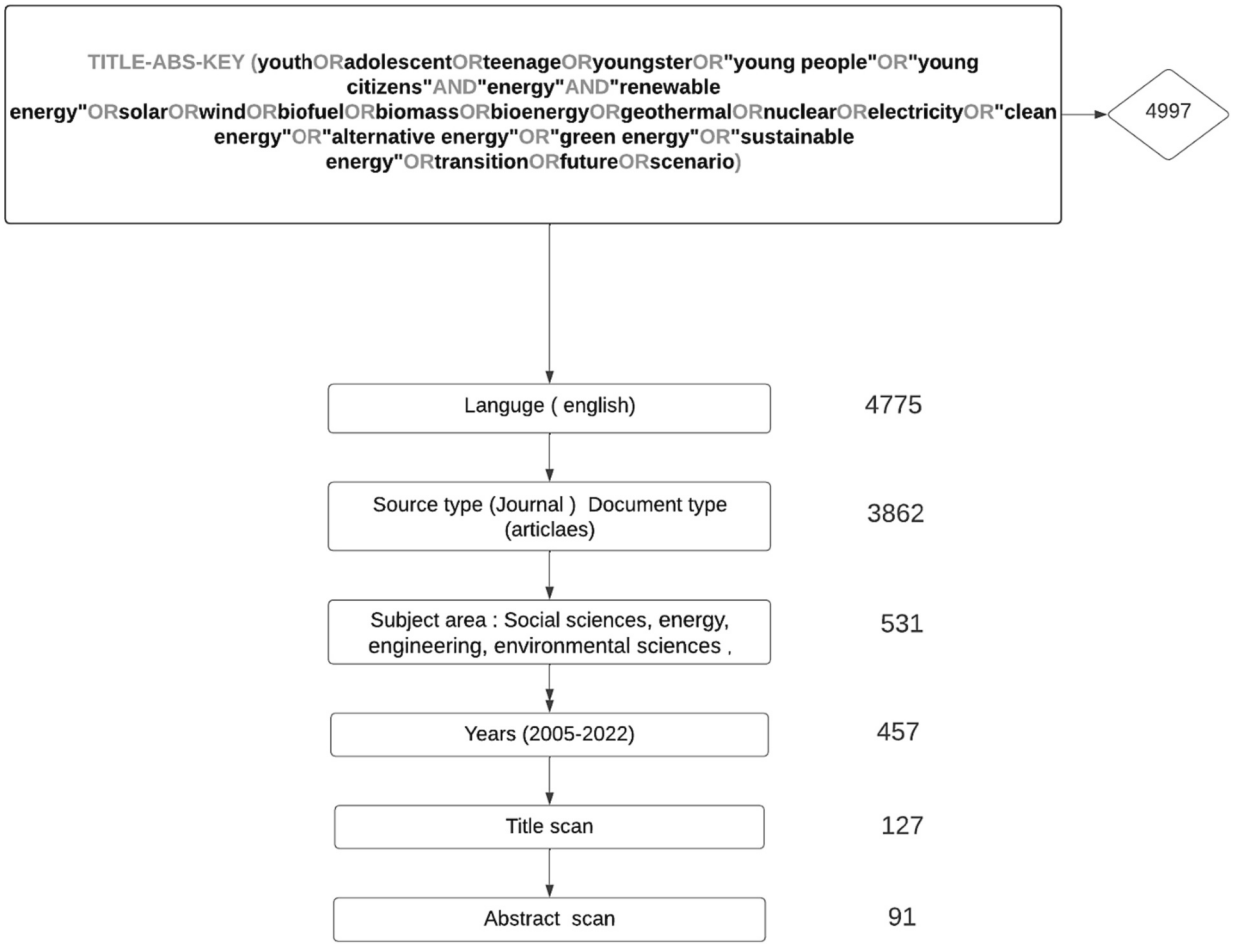

Fig. 1. Stages for searching and screening for publications related to youth and energy transition and the number of publications yielded from each stage.

based on the major topics addressed, and it was possible that an article could address multiple topics and thus contain multiple codes. For example, an article may explore both youth literacy and youth perceptions of renewable energy technologies. The complete set of codes across all papers was then reviewed and grouped and re-grouped into parent topics or themes through an iterative process that involved multiple rounds of coding until no new themes emerged and until all topics were assigned to a theme. An initial five rounds of coding were done by the lead author, which were subsequently reviewed and validated independently by the second author, followed by an additional two rounds of coding by the lead author to arrive at the final set of themes.

There are limitations to this approach and to this type of literature analysis and coding in general [47], for example limitations owing to the scope of search terms, the diversity of scholarship that may exist outside the peer reviewed journal literature (e.g., books, book chapters, technical reports), and the subjectivity in any qualitative approach to coding and thematic analysis. However, as noted by Leonhardt et al. [44], the approach adopted is commonly used by energy and social science scholars to explore or assess the state of knowledge and, arguably, provides a replicable method to enable the identification of dominant lines of inquiry and potential and critical gaps in scholarship.

4. Results

Search parameters covered 2005 to 2023, but results generated no papers meeting the search criteria until 2009 (Fig. 2). Results show a general increase in publications addressing youth and renewable energy over time, with a marked increase post-2019. The contributions of the last four years comprise 54 % of publications; 72 % of papers appeared after the adoption of the UN Agenda 2030. This period was marked by the global youth climate movement led by Greta Thunberg and a substantial international focus on youth mobilization, often focusing on disinvestment in fossil fuels and transitioning to renewable energy solutions. Included among such international initiatives are IRENA's 2020 and 2021 youth forums, The United Nations Industrial Development Organization (UNIDO) Vienna Energy Forum 2021, UN SEforALL and SDG7[1] Youth Constituency Summit 2021, and the SDG7 Youth Constituency of IEA's 2021 Global Commission on People-Centred Clean Energy Transitions [48]. Consistent with findings emerging from literature analyses in other fields of energy and social science research [44], Asia and Europe led in the geographic focus or concentration of research. Most articles focused on wind, solar, and bioenergy technologies. Not all articles specified what constitutes *youth*, but among those that did the focus was broad focusing on individuals aged from 10 to 40 years.

Five themes emerged from the analysis (Table 1), capturing how scholars approach or frame youth in energy transitions, namely: energy education, youth perceptions and attitudes about renewable energy, youth energy conservation behaviours and use, climate advocacy, and the role of youth in energy governance. Youth education is the dominant theme, addressed in some capacity in 57 % of all papers and including topics such as the importance of youth energy literacy, science-based educational programming, and education interventions. This was

[1] SDG7: The seventh goal of the 17 Sustainable Development Goals listed in Agenda 30 of the UN, which calls for 'affordable, reliable, sustainable and modern energy for all'.

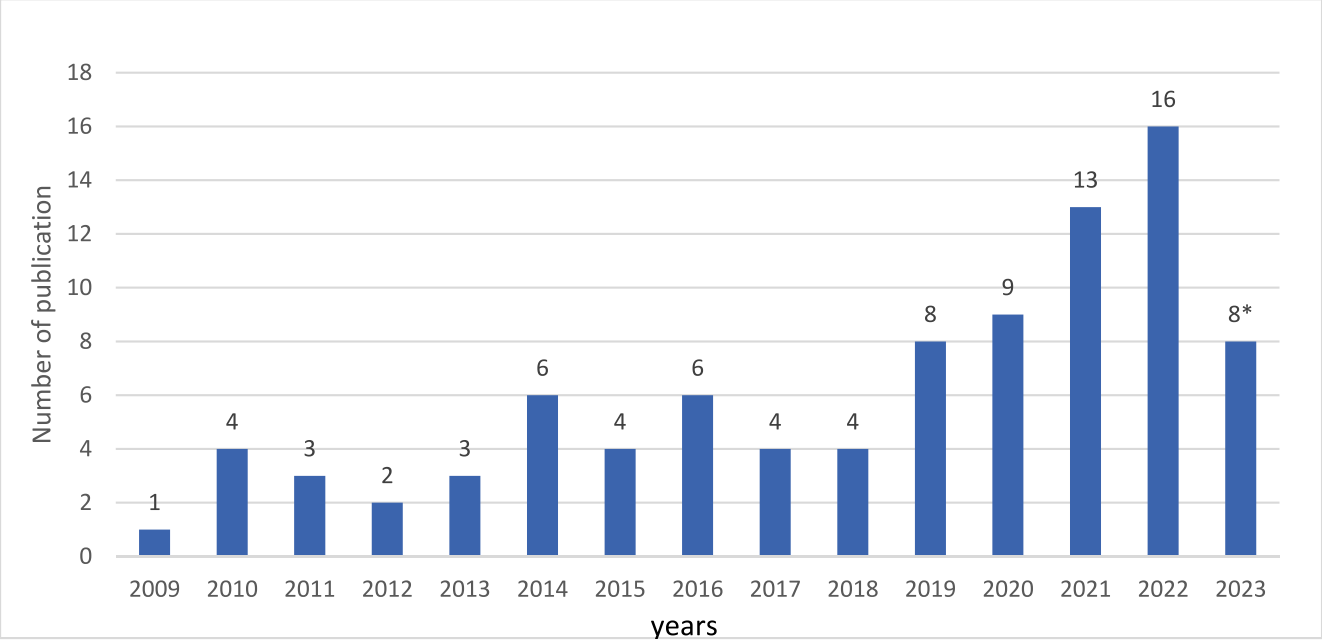

Fig. 2. Annual distribution of publications between 2005 and 2023 (* indicates partial year search results to 31 October 2023; the first paper to meet the criteria was published in 2009).

Table 1
Themes emerging from coding process and the percentage of papers addressing each theme.

Theme	Primary topic(s)	# (%) papers addressing the topic/ theme[a]	
Energy education	Importance of youth energy literacy [e.g., [52,55,57]	23	52 (57 %)
	Educational interventions for youth [e.g., 58,59,60]	15	
	Science-based youth educational programming [e.g., 61,62,63]	14	
Perceptions & attitudes	Youth perceptions and attitudes about renewable energy, including energy technologies [e.g., 64,65,66]	37	37 (41 %)
Energy conservation	Youth energy conservation behaviour and intent [e.g., 67,68,69]	24	24 (26 %)
Climate advocacy	Youth climate action and engagement in the multi-lateral energy space [e.g., 48,70,71]	21	21 (23 %)
Energy governance	Youth involvement and influence in local energy initiatives [e.g., 72,73,74]	9	9 (10 %)

[a] Total is greater than n = 91 (100 %) as papers often address multiple topics and/or cut across themes.

followed by papers exploring the perceptions and behaviours of youth regarding renewable energy and different energy technologies (41 %), youth energy conservation behaviours (26 %), and climate advocacy (23 %). Energy governance, specifically the role of youth in local energy initiatives and their influence on policy, was the least explored topic, addressed in only 10 % of papers.

Many of these topics, of course, are not mutually exclusive. For example, papers focused on youth perceptions about renewable energy or climate advocacy, often also addressed energy literacy or science-based education [49–51]. This was not surprising as energy education research can be cross-cutting of energy perceptions [49], conservation behaviours and climate advocacy [52]; while research focused on perceptions and attitudes about energy, including resistance to technologies, often refer to energy education and interventions as a solution [53]. The following sections address the state of research in each of these thematic areas and identify common topics and gaps in scholarship relevant to the role of youth in energy transition.

4.1. Energy education

Energy education emerged as a dominant research focus. Emphasis was predominately on three topics: youth energy literacy, the role and importance of educational interventions, and *science*-based energy and climate educational programs.

4.1.1. Importance of youth energy literacy

Energy literacy among youth was the most prevalent topic in the energy education theme, explored to some extent in 23 of 52 papers (Table 1). Lowan-Trudeau and Fowler [52] characterize energy literacy as multi-, inter- and transdisciplinary, and inherently about understanding the social, environmental, political and economic challenges and benefits and impacts of energy sources, projects, and technologies. Exploring the complexities of energy literacy among youth, Gladwin et al. [57] emphasize the importance of research that explores the "socio-cultural aspects of energy literacy as an underlying framework for energy and climate justice" (p. 251). Youth energy literacy was thus rarely a stand-alone topic, and typically was paired with topics such as energy conservation behaviour [75], youth response to the climate crisis [52], and political influence in youth energy literacy programming [76,147]. For example, Eaton and Day's [76] work on energy and climate change topics promoted in public education systems in western Canada show the influence of the fossil fuel industry in the classroom, whereby curricula are focused on individual responsibility for climate change while deflecting attention from one the largest sources of emissions in the region – the coal and oil sands production. Gladwin et al. [57] suggest that framing energy literacy only on individual responsibilities poses barriers to broader-scale transition and decarbonization efforts. Lowan-Trudeau and Fowler [52] thus argue the need for *critical energy literacy*, which encompasses all aspects of energy, environment, and climate in socio-cultural, geographical, and ecological contexts, emphasizing the importance of climate, environment, sustainability, gender and marginalized peoples' inequalities in relationship to energy initiatives [52,57].

In other instances, research focused on the mutually reinforcing

relationship between youth energy literacy and energy security, emphasizing that not only is youth energy literacy essential to energy security [60,77] but also that energy security can enhance youth capacity to engage meaningfully in energy transitions [78,79]. For example, renewable energy projects in rural communities can improve energy security, which, in turn, can lead to improved socio-economic and educational opportunities for young people, while mitigating the disproportionate and intergenerational effects of climate change on youth [78,80,81]. Faheem [82] and Stanczyk [48], for example, emphasize that just energy transitions are those that enable youth to discover their potential, build new skills, and become meaningful actors in energy transition [83].

Finally, albeit in only a few papers, energy literacy is explored in the context of promoting youth-led innovation, especially in rural areas. Faheem [82] for example, describes the impact of youth social entrepreneurship on energy security in rural Kenya, referring to a local millennial-age engineer who received national awards for the development of solar lanterns from recycled materials. Faheem [82] describes how the benefits extended beyond energy access to creating local job opportunities for youth and enhancing educational programming for school children in energy-insecure communities. Similarly, in rural Indonesia, Ratriyana et al. [23] report on the importance of youth engagement in local energy projects, emphasizing the ability of youth to find technical solutions to energy challenges and arguing that youth can be "agents of social change and drivers of development" (p. 290). Stanczyk [48] and Lowan-Trudeau and Fowler [52] thus emphasize that literacy is important for fostering innovation and to advancing community renewable energy implementation.

4.1.2. Educational interventions

Fifteen papers focused on specific educational tools or interventions for the delivery of energy education and promoting energy literacy (Table 1), whether within formal or informal learning environments. Research focus is often on exploring or testing educational interventions to change youth behaviours and reduce energy consumption, or to raise their awareness about energy transitions [84–88]. The types of interventions explored are diverse, from e-Learning activities to student-based monitoring of electric meters. Several studies focus on testing the effects of parents, teachers, and social media on youth energy literacy and actions [50,58,89]. What appears to be an emerging field of research is the use of *gaming*, including the merit of video gaming as a pedagogical tool [60,90]. Two papers in our sample focused on gaming as a mode of intervention and learning among youth to build energy literacy and solve energy challenges. For example, Ouariachi and Elving [60] report on the "We-Energy Game" for post-secondary youth – a game-based learning system whereby players are challenged to locate and employ local renewable energy solutions for their town or city to achieve energy neutrality. Ouariachi et al. [90] indicate that such energy game-based interventions not only build cognitive capacity but can also motivate youth to contribute meaningfully to local energy transitions in their communities.

4.1.3. Science-based educational programming for youth

Finally, the importance of science-based educational programming emerged in 14 of the papers examined (Table 1), specifically engineering and technical-based education and climate science. Cornelius et al. [89] and Aruta [91] indicate that science-based pedagogies have a significant role in building youth's cognitive understanding of climate change and energy problems, including the ability to frame future mitigation behaviours. Moreover, adapting science learning to real energy-life problems can manifest in energy citizenship and engagement with climate and energy initiatives [62,63,92]. Research often referred to different strategies such as after-school projects [62,63] and science camps or campaigns [93] that emphasize leadership, problem solving, and communication. Several examples from the literature focused on connecting curriculum with real world community energy problems to motivate students to develop solutions [61,63,64,92]. Wensing et al. [94] and Mori and Tasaki [95] refer to the Core Curriculum for Civic Engagement (CC4CE) initiative, a teaching-learning program that links schools with communities to engage students with social and environmental problems worldwide.

4.2. Perceptions and attitudes

A total of 37 papers (41 %) focused on testing or exploring youth awareness about renewable energy and youth perspectives and attitudes about renewables (Table 1). The primary focus is on youth awareness of and views toward specific technologies, namely wind, solar, and bioenergy [49,65,96–98,149]; understanding youth perspectives on zero-carbon economies [66,148,150]; and youth awareness about energy saving actions [51,67,99]. The research is typically set in the context of climate change mitigation. Halder and colleagues, for example, have made numerous contributions to studies of youth knowledge, awareness, and attitudes toward renewables, especially bioenergy in Finland [49,64,96,97,100], and several other studies focus on youth attitudes toward climate change and renewable energy sources across different socio-economic contexts [99,101]. Youth are also depicted by scholars to be more willing than older populations to accept, use, and financially support renewable energy technologies [102–104], with some exception among youth in rural areas [105,106]. Youth acceptance of renewable energy is often associated with a higher awareness of renewable energy benefits [102] due to higher exposure to social media, energy education, and renewable energy campaigns [107].

Included among the rationales offered in the literature for the dominant focus on testing or understanding youth perceptions and awareness of renewables is that youth, especially in rural regions or developing nations, comprise a large portion of the population and understanding perceptions and attitudes is essential to shaping youth to be sustainable energy consumers and adopters of renewable energy technologies [64,65]. Similar rationales for the emphasis placed on understanding youth energy perceptions and awareness are offered by Assali et al. [98], Akinwale [51], and Avramides et al. [50], explaining that youth are current and future consumers of energy and ultimately future energy decision-makers [65].

4.3. Energy conservation

Research focused on energy conservation, specifically youth energy conservation behaviours and intent, comprises 26 % of our sample (Table 1). Youth's energy conservation behaviour is the focal point of education and educational interventions that seek to change youth behaviour toward more sustainable actions [58,108]. Energy conservation is expressed in multiple ways in the papers sampled, such as energy saving actions [51], green energy use [109], and energy efficient technologies [68]. The rationales presented for studying youth's energy-saving behaviours are based on the notion that youth are intensive energy consumers [68], victims of the climate crises, and must also be part of the climate solution [87,110]. Youth as energy intensive consumers, notwithstanding being more climate-aware, is attributed to high use of individual technologies (e.g., smart phones, tablets, gaming systems) [69,111]– a phenomenon referred to as the Nintendo-effect [112]. However, scholars also note that educational systems have a significant role to play in changing youth energy consumption behaviours by realigning their energy consumption lifestyles to better match their environmental values, norms, and beliefs [54,64,65,68,113]. As Schmidt et al. [114] explain, youth can be agents of change by influencing members in their social networks with energy-saving behaviour.

4.4. Climate advocacy

Climate change intersected with all topics identified in our review, for example, in relation to youth energy education [55], perceptions

about low-carbon futures [66], and individual pro-environmental behaviours [91] In these contexts, youth are often referred to, explicitly or implicitly, as influential energy citizens; however, as Jorgenson et al. [55] emphasize, the influence of youth on large-scale change is possible only when youth are involved in collective actions. Thus, not surprisingly, climate advocacy emerged as a core theme in the energy scholarship, addressed in 21 % of papers (Table 1). A small portion of this literature emphasized formal litigated actions among youth as an instrument to influence energy transitions toward a low-carbon future. Olson [115], for example, in the US context, refers to litigation as an instrument for youth under the voting age to force governments to mandate policies for emissions reduction by citing court cases where youth have claimed that current policies (climate, energy, emissions) are practicing a form of discrimination against future generations and violate their rights.

Most of the research captured by this theme is focused on the increasing role of youth in the multi-lateral energy space. The multi-lateral energy space is where actors (e.g., NGOs, women, Indigenous communities, labour unions, academia, businesses, local governments) come together to discuss and work toward coordinated and collaborative actions of global importance [116]. Authors often refer to the Youth Strike for Climate (YSC), sparked by the Swedish teenager Greta Thunberg in 2018, which motivated youth worldwide to engage in climate action [52], giving rise to other advocacy and grassroots initiatives to protest "unjust climate policies" that threaten future generations [80,118,117]. These advocacy groups range from regional to national university-based groups, such as Student Energy in Canada, contributing to energy literacy and youth empowerment through capacity building, knowledge dissemination, and advocacy actions for climate and energy justice [52]; to multi-lateral initiatives, such as the Youth Sustainable Energy Hub – the first global platform that aims to unite youth engagement in sustainable energy, developed by SDG7 Youth Constituency, and Brazil, Russia, India, China and South Africa (BRICS) Youth Energy Agency (YAE) [48].

That said, the literature also identifies two core challenges associated with youth climate advocacy. The first concerns the *actual* influence of youth councils or advocacy groups in the multi-lateral energy space, specifically international forums. Stanczyk [48] notes that while such youth councils are perceived to be effective in building relationships, providing quality input, and facilitating access to knowledge and resources among youth, multi-lateral spaces are not always inclusive, especially for marginalized and vulnerable populations. Strzelecki [71] implies that joining multilateral events does not necessarily translate to direct influence on climate and energy policies or actions – often owing to lack of clarity on *how* youth advocacy is channelled to decisions makers, the limited opportunity for more nuanced national to regional-focused conversations, and the limited long-term commitment beyond the specific events themselves. The second challenge concerns the conflicts or inequalities that emerge from climate movements; for example, rural and Indigenous citizens shouldering the potential negative externalities (e.g., wind turbines and landscapes) connected to renewable energy [119], and gender inequalities reflected by fossil fuel industries toward female climate activists [52]. Adding to the above are the intergenerational inequalities raised by climate movement discourses [118] and the critiques of past generations' actions and decisions and the effects on the climate crises of today.

4.5. Energy governance

Bonsu et al. [119] argue that youth are legitimate actors that must be engaged in the localization of global SDGS – including SDG7 (Affordable and Clean Energy). Yet, the final theme emerging from our results, addressed in the least number of papers, is the role of youth in, and their influence on, local energy initiatives (Table 1). Only nine papers were identified that focused specifically on the role of youth in energy governance at the local level, notwithstanding a diversity of literature suggesting that collective engagement at the community level is essential to community renewable energy transitions [120–124,148]. Surprisingly, very few papers explore specific methods or approaches to youth engagement in energy decision-making at the local level; those that do tend to emphasize citizens' assemblies and forums, online opinion polls, and preference survey instruments on governmental platforms [72,73]. However, as Dunlop et al. [72] note, such methods are often ineffective for empowering youth.

There are barriers identified in research that prevent youth from being engaged in energy governance or decision-making at the local level. One of these barriers is the lack of recognition of youth and their exclusion from voting and consultation processes. For example, Dunlop et al. [72] report on the disproportionate impacts of hydraulic fracking on young people in England (i.e., longer-term impacts, operations adjacent to schools), but note that youth under the legal voting age are excluded from government's formal consultation and decision-making processes. A second barrier identified is observed in rural and traditional tribal communities in Indonesia, where the values, norms, and beliefs of elder leaders may limit the voice and role of youth, preventing their social or political engagement in local renewable energy initiatives at the community level [23]. Other researchers emphasize men's power over youth (and women) when making decisions about local energy initiatives. For example, in Germany, Radtke and Ohlhorst [74] report patterns of injustice in community renewable energy projects, describing community energy as an "exclusive club for men". Finally, in the absence of larger regional to national organized efforts, Nathan et al. [77] report that at the local level youth often lack the capacity to engage in rural energy projects – a scenario which, based on experiences with microgrid initiatives in rural India, can lead to project dysfunction and failure.

5. Discussion

Five themes emerged from the results that characterize the focus of recent scholarship on youth in energy transitions (Table 1). The dominant focus is on the importance of youth energy literacy and educating youth about energy options and the relationship between energy and environment [52,55]. Much of this literature emphasizes the importance of knowledge and capacity building among youth. This was followed by scholarship that explored and often tested youth knowledge about and perspectives on energy, typically energy options or different energy technologies [49,66]. Literature focused on youth energy conservation behaviour and climate advocacy captured equal attention in the sample of papers reviewed. Studies on youth energy conservation behaviours often focus on youth lifestyles and energy consumption [51,98], especially in developing regions, based on the premise that youths' understanding of energy conservation will shape whether and how we achieve meaningful emissions reduction and energy transitions [50]. The climate advocacy literature focused on climate movements and youth participation in multilateral energy and climate spaces [48,71], emphasizing the importance of youth collective actions to drive energy and climate action. Finally, results show limited scholarship has focused on youth in energy governance, specifically the role and influence of youth at the local level in on-the-ground energy initiatives. Based on our analysis and the topics and themes that emerged, we offer three observations concerning the current scope of scholarship and the framing of youth in energy transitions. Each of these is discussed below along with recommendations for future scholarship to support the societal embedding of energy transitions through youth engagement.

5.1. Science-based education vs. critical energy literacy

Energy education and literacy are essential to building the required agency for youth to be key actors in energy transitions. Bastien and Holmarsdottir [32] argue that quality education and training are foundational to youth driven innovation for social change. However,

research related to energy education and literacy is focused primarily on science-based education (e.g., technology, environmental science, engineering), with the main objective often to teach youth about energy technologies, climate, and conservation [62,63,92]. Although important, research on energy literacy must also consider the social, political, and cultural contexts in which such educational programs or interventions are occurring. We recommend that youth-based energy literacy extend beyond the traditional science disciplines to better position youth as next-generation energy leaders [25] who are "capable of leading and making a difference" and, ultimately, reshaping power structures [143]. Thus, building energy literacy among youth must consider multi-, inter-, and trans-disciplinary curricula that address energy literacy as multi-dimensional relationships between energy technology, social systems, cultures, and political systems. This is what Lowan-Trudeau and Fowler refer to as "critical energy literacy" [52,57], which can ultimately help shape more equitable energy futures, support the long-term sustainability of energy transitions, and provide for more an inclusive energy planning and policy environment.

Related to the development of critical energy literacy among youth is the need for decolonization of energy literacy. Various scholars emphasize the importance of Indigenous knowledge and values in transitions research, but at the same time indicate its underrepresentation in energy transitions scholarship [145,144,146]. Rakshit et al. [147] report that energy transitions research in the Indigenous context in general is limited to only a few studies focused on energy planning and renewables. Our analysis of scholarship on youth energy literacy reveals limited attention to different knowledge systems and world views among youth, especially Indigenous youth, and how such beliefs and values may be considered in, and inform, youth-based energy literacy programs. This is a substantial gap in both energy transitions scholarship and energy literacy, considering that in many instances the greatest energy transition potentials are in rural regions of the developing south and across the Arctic – regions with large Indigenous youth populations. Remote Indigenous communities have shown resistance to unfamiliar energy technologies [146], emphasizing the importance of energy literacy that reflects cultural systems, knowledge, and traditions. Youth Indigenous knowledge and culture is absent in recent international scholarship focused on youth education and energy literacy, yet braiding of Indigenous youth knowledge and culture with Western science is essential to informing just energy transitions.

5.2. Perceptions and behaviour vs. pathways for youth influence

There is a tendency in the literature to focus on examining youth perceptions of renewable energy or energy technologies and how youth use and consume energy. Although such literature provides important insight to youth awareness of renewable energy and their likelihood to choose renewable energy options, and youth energy conservation behaviour, we observed limited direction on whether or how youth awareness about renewables and their and energy choices shape or influence youth engagement as core actors in energy transition initiatives or renewables deployment. Scholars often refer to youth as agents of change in energy and sustainability transitions [72,119]. However, understanding the role and influence of youth in local to global energy transitions requires much more than exploring how youth perceive and consume energy [55,151]; it requires identifying, understanding, and creating the enabling conditions and environments to translate youth awareness, perceptions, or values of energy into collective actions that can open windows for energy transitions.

Scholars of energy citizenship emphasize that a dominant focus on youth awareness and individual choices wrongfully places the responsibility of transition and climate change mitigation onto individual citizens [7,76], which does not align with conceptualizations of youth as a niche in driving innovation and change in local to global energy regimes. We agree that it is important to understand youth awareness of renewable energy and to inform sustainable energy choices, but this understanding alone is not enough to empower energy transitions; research is needed to bridge the gap between perceptions and behaviour and support the pathways by which youth can be meaningfully engaged in energy transitions. This requires a better understanding of youth's values regarding their vision of a renewables-based society, and developing new narratives of energy futures in education, policy, and planning that reflect youth's values [55] and support their long-term engagement with energy transitions.

5.3. Advocacy and multi-lateral initiatives vs. engagement and influence at the community level

Youth are increasingly engaged in collective actions to challenge national and international energy and climate regimes [115,125], as evidenced by global youth climate movements and engagement in national to international forms such as the UN, IRENA, and IEA, to name a few. Such forums present important opportunities for youth from diverse backgrounds, ethnicities, education, and experiences to organize and communicate a united vision, to learn from each other, and to express their values and priorities regarding energy and climate futures in formal policy settings [71]. However, the extent to which such youth organizations and forums operating in multilateral spaces influence energy policies or transition trajectories has received much less attention [48]. A critical review of the youth programs offered and supported by intergovernmental organizations and energy agencies is needed to ensure their inclusiveness and effectiveness [71], coupled with fundamental research focused on youth's immediate and longer-term goals, aspirations, and expectations about their engagement in the multilateral energy space.

Although youth engagement in the multi-lateral energy space is important, the dominant focus on youth action in this sphere may have resulted in an inadequate body of research focused on the roles of youth in shaping local energy initiatives at the community level. Notwithstanding scholarship pointing to community renewable energy as foundational to energy transition [44,126], alongside international recognition of youth as important actors in energy and climate futures [24], we observed very little research focused on youth engagement in energy transitions or renewable energy projects at the regional to community scale. Community energy scholarship often addresses local government actors and community energy initiatives [77], which are often dominated by older male participants [23,74]. Nelson [127] emphasizes the importance of embracing generational leadership at the local level, meaning that youth must be active participants in community energy initiatives and empowered to drive change [128,129,148]. McMaster et al. [25] characterize youth as the next generation leaders of local energy in rural and remote regions, and thus foundational to long-term energy transition success. Yet, consistent with our findings, recent scholarship on advancing community energy does not necessarily align with the nuances of rural, remote, and especially Indigenous contexts [25]. The limited attention to youth in shaping energy transitions at the local level, especially in rural and remote contexts, is a critical gap in energy scholarship. Research and development are needed to strength opportunities for youth leadership, creating programs and pathways for engaging youth in energy transition – from promoting energy conservation behaviours to formal community energy planning. Consistent with Jorgenson et al. [55], youth must be engaged as innovators within a broader community and social network of energy actors.

6. Conclusions

This paper set out to explore how youth are conceptualized in energy transitions scholarship – whether as influential energy citizens or merely passive actors. More than half of the papers identified in our sample that explore youth in energy transitions were published in the last four years, and nearly three-quarters were published post-adoption of UN Agenda 2030. The majority of papers did not explore youth as energy citizens *per*

se; rather, the dominant focus is on the importance of youth energy education or literacy, testing youth's perceptions about renewable energy technologies, and exploring youth's behaviour as energy consumers. Only limited research was observed focused on youth as energy citizens to empower transitions. As such, there is a need for youth energy scholarship at two complimentary scales. First, research is needed at the national to international scale, focused on the rationale, value added and influence of youth engagement in the multi-lateral energy space. Notwithstanding increasing international youth forums, largely focused on climate or energy and climate interaction, there is limited understanding of the drivers or expectations of youth when engaging in such spaces, and the impact or influence of such national to international youth forums and organizations on key decisions, policies, or other actions or innovations that shape energy regimes. Second, research is needed at the local or community scale and application to enable youth's roles in local energy planning and renewable energy projects; this is especially important in rural and remote regions, where youth must play a more prominent role in shaping energy projects and energy futures [25]. Although multiple opportunities exist to improve youth energy literacy and explore youth energy perspectives and behaviours, research focused on youth as energy citizens in governance from the national to the local is essential to empowering youth and providing a foundation to support long-term, sustainable energy transitions.

CRediT authorship contribution statement

Arwa Jaradat: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft. **Bram Noble:** Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Validation, Writing – review & editing. **Greg Poelzer:** Funding acquisition, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This research was funded by the Social Sciences and Humanities Research Council of Canada, Grant no. 895-2019-1007. We wish to acknowledge the Community Appropriate Sustainable Energy Security (CASES) partnership for supporting this work.

References

[1] A. Cherp, V. Vinichenko, J. Jelwee, E. Brutschin, B. Sovacool, Integrating techno-economic, socio-technical and political perspectives on national energy transitions: a meta-theoretical framework, Energy Res. Soc. Sci. 37 (2018) 175–190, https://doi.org/10.1016/j.erss.2017.09.015.
[2] J. Markard, R. Raven, B. Truffer, Sustainability transitions: an emerging field of research and its prospects, Res. Policy 41 (6) (2012), https://doi.org/10.1016/j.respol.2012.02.013.
[3] F.W. Geels, Technological transitions as evolutionary reconfiguration processes: a multi-level perspective and a case-study, Res. Policy 31 (8–9) (2002), https://doi.org/10.1016/S0048-7333(02)00062-8.
[4] G. Verbong, F. Geels, The ongoing energy transition: lessons from a socio-technical, multi-level analysis of the Dutch electricity system (1960–2004), Energy Policy 35 (2) (2007), https://doi.org/10.1016/j.enpol.2006.02.010.
[5] M. Lawhon, J. Murphy, Socio-technical regimes and sustainability transitions: insights from political ecology, Prog. Hum. Geogr. 36 (3) (2012), https://doi.org/10.1177/0309132511427960.
[6] S.R. Jansma, L.A.N. Long, D. Lee, Understanding energy citizenship: how cultural capital shapes the energy transition, Energies 16 (5) (2023) 2106, https://doi.org/10.3390/en16052106.
[7] M. Wahlund, J. Palm, The role of energy democracy and energy citizenship for participatory energy transitions: a comprehensive review, Energy Res. Soc. Sci. 87 (2022), 102482, https://doi.org/10.1016/j.erss.2021.102482.
[8] M. Ryghaug, T.M. Skjølsvold, S. Heidenreich, Creating energy citizenship through material participation, Soc. Stud. Sci. 48 (2) (2018) 283–303, https://doi.org/10.1177/0306312718770286.
[9] P. Devine-Wright, Energy citizenship: psychological aspects of evolution in sustainable energy technologies, in: J. Murphy (Ed.), Governing Technology for Sustainability, Earthscan, London, 2007, pp. 63–88.
[10] G. Mullally, N. Dunphy, P. O'Connor, Participative environmental policy integration in the Irish energy sector, Environ. Sci. Policy 83 (2018) 71–78, https://doi.org/10.1016/j.envsci.2018.02.007.
[11] M.J. Burke, J.C. Stephens, Energy democracy: goals and policy instruments for sociotechnical transitions, Energy Res. Soc. Sci. 33 (2017) 35–48, https://doi.org/10.1016/j.erss.2017.09.024.
[12] B. van Veelen, W. Eadson, Assembling community energy democracies, Voluntary Sector Rev. 11 (2) (2020) 231–249, https://doi.org/10.1332/204080519X15740562779512.
[13] M.P.C. Weijnen, Z. Lukszo, S. Farahani (Eds.), Shaping an Inclusive Energy Transition, Springer International Publishing, 2021, https://doi.org/10.1007/978-3-030-74586-8.
[14] B. Pel, A. Debourdeau, R. Kemp, A. Dumitru, E. Vadovics, M. Schäfer, M. Markantoni, B. Schmid, F. Fahy, A. Fransolet, K. Thalberg, Energy Citizenship: Ideals, Ideology and Ideal-types in the Energy Transition. European Forum for Studies of Policies for Research and Innovation (EU-SPRI) (1–3) June, 2022 (Utrecht (NL)).
[15] T. Ringholm, Energy citizens – conveyors of changing democratic institutions? Cities 126 (2022), 103678 https://doi.org/10.1016/j.cities.2022.103678.
[16] R.R. Brown, M.A. Farrelly, D.A. Loorbach, Actors working the institutions in sustainability transitions: the case of Melbourne's stormwater management, Glob. Environ. Chang. 23 (4) (2013), https://doi.org/10.1016/j.gloenvcha.2013.02.013.
[17] D. Loorbach, Transition management for sustainable development: a prescriptive, complexity-based governance framework, Governance 23 (1) (2010) 161–183, https://doi.org/10.1111/j.1468-0491.2009.01471.x.
[18] B. Lennon, N. Dunphy, C. Gaffney, A. Revez, G. Mullally, P. O'Connor, Citizen or consumer? Reconsidering energy citizenship, J. Environ. Policy Plan. 22 (2) (2020) 184–197, https://doi.org/10.1080/1523908X.2019.1680277.
[19] B. Fischer, G. Gutsche, H. Wetzel, Who wants to get involved? Determining citizen willingness to participate in German renewable energy cooperatives, Energy Res. Soc. Sci. 76 (2021), 102013, https://doi.org/10.1016/j.erss.2021.102013.
[20] C. Walker, G. Poelzer, R. Leonhardt, B. Noble, C. Hoicka, COPs and 'robbers?' Better understanding community energy and toward a communities of place then interest approach, Energy Res. Soc. Sci. 92 (2022), 102797, https://doi.org/10.1016/j.erss.2022.102797.
[21] N. DellaValle, V. Czako, Empowering energy citizenship among the energy poor, Energy Res. Soc. Sci. 89 (2022), 102654, https://doi.org/10.1016/j.erss.2022.102654.
[22] A. Silvast, G. Valkenburg, Energy citizenship: a critical perspective, Energy Res. Soc. Sci. 98 (2023), 102995, https://doi.org/10.1016/j.erss.2023.102995.
[23] I.N. Ratriyana, L.D. Setiawan, G.A. Yudarwati, Development communication for youth empowerment in Indonesia's renewable energy projects, Asian J. Commun. 32 (4) (2022) 290–308, https://doi.org/10.1080/01292986.2021.2007273.
[24] Z. Zhongming, L. Linong, Y. Xiaona, Z. Wangqiang, L. Wei, Global Youth Call for a Just Energy Transition at IRENA Eleventh Assembly, 2021.
[25] R. McMaster, B. Noble, G. Poelzer, V. Menghwani, Local capacity for energy transition in northern and indigenous communities: analysis of Gwich'in communities in Northwest Territories, Canada, Arctic 76 (2) (2023) 112–243, https://doi.org/10.14430/arctic77183.
[26] M. Bucholtz, Youth and cultural practice, Annu. Rev. Anthropol. 31 (1) (2002) 525–552.
[27] D. Garriguet, Portrait of youth in Canada: Data report Chapter 1: Health of youth in Canada, Statistics Canada, 2021. https://www150.statcan.gc.ca.
[28] United Nation, Department of Economic and Social Affairs. The sustainable development goals report 2020, Available on, https://unstats.un.org/sdgs/report/2020/The-Sustainable-Development-Goals-Report-2020.pdf. (Accessed 10 March 2023).
[29] United Nations, World youth report: youth and the 2030 Agenda for Sustainable Development, Available at:, New York, UN, 2018. https://www.un.org/development/desa/youth/world-youth-report/wyr2018.html.
[30] A. Khan, A. Javed, S. Batool, F. Hussain, H. Mahmood, V. Ahmed, The role of youth in sustainable development, Available from, https://sdpi.org/sdpiweb/publications/files/commissioned-Paper-The-role-of-youth-in-sustainable-development-perspectivesfromSouthAsia.pdf, 2016.
[31] P. Solís, B. McCusker, N. Menkiti, N. Cowan, C. Blevins, Engaging global youth in participatory spatial data creation for the UN sustainable development goals: the case of open mapping for malaria prevention, Appl. Geogr. 98 (2018) 143–155, https://doi.org/10.1016/j. apgeog.2018.07.013.
[32] S. Bastien, H.B. Holmarsdottir (Eds.), Youth as Architects of Social Change: Global Efforts to Advance Youth-driven Innovation, Springer International Publishing, 2017, https://doi.org/10.1007/978-3-319-66275-6.
[33] S. Mallick, M. Rahman, Empowering youth lasting change; SEED SCALE a process to youth engagement and sustainable development, Technium Soc. Sci. J. 13 (2020) 501.

[34] A. Bandura, L. Cherry, Enlisting the power of youth for climate change, Am. Psychol. 75 (7) (2020) 945–951, https://doi.org/10.1037/amp0000512.
[35] K.C. Busch, N. Ardoin, D. Gruehn, K. Stevenson, Exploring a theoretical model of climate change action for youth, Int. J. Sci. Educ. 41 (17) (2019) 2389–2409, https://doi.org/10.1080/095 00693.2019.1680903.
[36] H. Han, S.W. Ahn, Youth mobilization to stop global climate change: narratives and impact, Sustainability 12 (10) (2020) 4127, https://doi.org/10.3390/su12104127.
[37] B.S. Ballew, Elsevier's Scopus® database, J. Electronic Resour. Med. Libr. 6 (3) (2009) 245–252, https://doi.org/10.1080/15424060903167252.
[38] J.F. Burnham, Scopus database: a review, Biomed. Digit. Libr. 3 (1) (2006) 1, https://doi.org/10.1186/1742-5581-3-1.
[39] C. López-Illescas, F. de Moya-Anegón, H.F. Moed, Coverage and citation impact of oncological journals in the Web of Science and Scopus, J. Informet. 2 (4) (2008) 304–316, https://doi.org/10.1016/j.joi.2008.08.001.
[40] S. Baykoucheva, Selecting a database for drug literature retrieval: a comparison of MEDLINE, Scopus, and Web of Science, Sci. Technol. Libr. 29 (4) (2010) 276–288, https://doi.org/10.1080/0194262X.2010.522946.
[41] P. Mongeon, A. Paul-Hus, The journal coverage of Web of Science and Scopus: a comparative analysis, Scientometrics 106 (1) (2016) 213–228, https://doi.org/10.1007/s11192-015-1765-5.
[42] M. Šember, A. Utrobičić, J. Petrak, Croatian medical journal citation score in Web of Science, Scopus, and Google Scholar, Croat. Med. J. 51 (2) (2010) 99–103, https://doi.org/10.3325/cmj.2010.51.99.
[43] E.M.L. Bergman, Finding citations to social work literature: The relative benefits of using Web of Science, Scopus, or Google Scholar, J. Academic Librarianship 38 (6) (2012) 370–379, https://doi.org/10.1016/j.acalib.2012.08.002.
[44] R. Leonhardt, B. Noble, G. Poelzer, P. Fitzpatrick, K. Belcher, G. Holdmann, Advancing local energy transitions: a global review of government instruments supporting community energy, Energy Res. Soc. Sci. 83 (2022), 102350, https://doi.org/10.1016/j.erss.2021.102350.
[45] J. Attridge-Stirling, Thematic networks: an analytic tool for qualitative research, Qual. Res. 1 (3) (2001) 385–405, https://doi.org/10.1177/146879410100100307.
[46] A. Morrison, C.J. Westbrook, B.F. Noble, A review of the flood risk management governance and resilience literature: flood risk management governance and resilience literature, J. Flood Risk Manage. 11 (3) (2018) 291–304, https://doi.org/10.1111/jfr3.12315.
[47] B.K. Sovacool, J. Axsen, S. Sorrell, Promoting novelty, rigor, and style in energy social science: towards codes of practice for appropriate methods and research design, Energy Res. Soc. Sci. 45 (2018) 12–42, https://doi.org/10.1016/j.erss.2018.07.007.
[48] A. Stanczyk, Getting to COP27: bridging generational divide, Development 65 (1) (2022) 42–47, https://doi.org/10.1057/s41301-022-00335-2.
[49] P. Halder, S. Havu-Nuutinen, J. Pietarinen, P. Pelkonen, Bio-energy and youth: analyzing the role of school, home, and media from the future policy perspectives, Appl. Energy 88 (4) (2011) 1233–1240, https://doi.org/10.1016/j.apenergy.2010.10.017.
[50] K. Avramides, B. Craft, R. Luckin, Modelling teenage personal contexts to support technology enhanced enquiry into personal energy consumption, Comput. Educ. 69 (2013) 377–386, https://doi.org/10.1016/j.compedu.2013.07.019.
[51] Y.O. Akinwale, Awareness and perceptions of university students in the business college towards energy savings and renewable energy sources in Saudi Arabia, Int. J. Energy Econ. Policy 12 (3) (2022) 470–476, https://doi.org/10.32479/ijeep.13004.
[52] G. Lowan-Trudeau, T.A. Fowler, Towards a theory of critical energy literacy: the Youth Strike for Climate, renewable energy and beyond, Aust. J. Environ. Educ. 38 (1) (2022) 58–68, https://doi.org/10.1017/aee.2021.15.
[53] Z. Szakály, P. Balogh, E. Kontor, Z. Gabnai, A. Bai, Attitude toward and awareness of renewable energy sources: Hungarian experience and special features, Energies 14 (1) (2020) 22, https://doi.org/10.3390/en14010022.
[54] G. Kaur, P. Gupta, M. Syal, Appraisal of knowledge of youth towards energy conservation and efficiency in india: An exploratory research, OIDA International Journal of Sustainable Development 7 (1) (2014) 11–16.
[55] S.N. Jorgenson, J.C. Stephens, B. White, Environmental education in transition: a critical review of recent research on climate change and energy education, J. Environ. Educ. 50 (3) (2019) 160–171, https://doi.org/10.1080/00958964.2019.1604478.
[56] R. McMaster, B. Noble, G. Poelzer, Assessing local capacity for community appropriate sustainable energy transitions in northern and remote Indigenous communities, Renew. Sust. Energ. Rev. 191 (2024), https://doi.org/10.1016/j.rser.2023.114232.
[57] D. Gladwin, C. Karsgaard, L. Shultz, Collaborative learning on energy justice: international youth perspectives on energy literacy and climate justice, J. Environ. Educ. 1–10 (2022), https://doi.org/10.1080/00958964.2022.2113019.
[58] M.J. Bekker, T.D. Cumming, N.K.P. Osborne, A.M. Bruining, J.I. McClean, L. S. Leland, Encountering electricity saving in a university residential hall through a combination of feedback, visual prompts, and incentives, J. Appl. Behav. Anal. 43 (2) (2010) 327–331, https://doi.org/10.1901/jaba.2010.43-327.
[59] T.B. Bell, N. Toth, L. Little, M.A. Smith, Planning to save the planet: using an online intervention based on implementation intentions to change adolescent self-reported energy-saving behavior, Environ. Behav. 48 (8) (2016) 1049–1072, https://doi.org/10.1177/0013916515583550.
[60] T. Ouariachi, V. Elving, Accelerating the energy transition through serious gaming: testing effects on awareness, knowledge and efficacy beliefs, Electron. J. E-Learn. 18 (5) (2020), https://doi.org/10.34190/JEL.18.5.004.
[61] A.C. Barton, E. Tan, We be burnin'! Agency, identity, and science learning, J. Learn. Sci. 19 (2) (2010) 187–229, https://doi.org/10.1080/10508400903530044.
[62] D. Birmingham, A. Calabrese Barton, Putting on a green carnival: youth taking educated action on socio-scientific issues: putting on a green carnival, J. Res. Sci. Teach. 51 (3) (2014) 286–314, https://doi.org/10.1002/tea.21127.
[63] M.E. Jordan, S. Zuiker, W. Wakefield, M. DeLaRosa, Real work with real consequences: enlisting community energy engineering as an approach to envisioning engineering in context, J. Pre-College Eng. Educ. Res. 11 (1) (2021), https://doi.org/10.7771/2157-9288.1294.
[64] P. Halder, P. Prokop, C.-Y. Chang, M. Usak, J. Pietarinen, S. Havu-Nuutinen, P. Pelkonen, M. Cakir, International survey on bioenergy knowledge, perceptions, and attitudes among young citizens, Bioenergy Res. 5 (1) (2012) 247–261, https://doi.org/10.1007/s12155-011-9121-y.
[65] M. Yazdanpanah, N. Komendantova, Z.N. Shirazi, J. Linnerooth-Bayer, Green or in between? Examining youth perceptions of renewable energy in Iran, Energy Res. Soc. Sci. 8 (2015) 78–85, https://doi.org/10.1016/j.erss.2015.04.011.
[66] B. Kaczmarczyk, I. Urych, Perception of the transition to a zero-emission economy in the opinion of polish students, Energies 15 (3) (2022) 1102, https://doi.org/10.3390/en15031102.
[67] N.S.M. Salleh, Youth awareness towards energy saving, Adv. Sci. Lett. 22 (10) (2016) 3004–3006, https://doi.org/10.1166/asl.2016.7119.
[68] A. Al Mamun, N. Hayat, M.M. Masud, Q. Yang, A.A. Salameh, M.F. Md Salleh, Energy conservation behaviour among the Malaysian youth: a study under the premises of value-belief-norm model, Front. Energy Res. 10 (2022), 902619, https://doi.org/10.3389/fenrg.2022.902619.
[69] T.H. Christensen, E. Rommes, Don't blame the youth: the social-institutional and material embeddedness of young people's energy-intensive use of information and communication technology, Energy Res. Soc. Sci. 49 (2019) 82–90, https://doi.org/10.1016/j.erss.2018.10.014.
[70] B. Nkrumah, Courting emissions: climate adjudication and South Africa's youth, Energy Sustain. Soc. 11 (1) (2021) 45, https://doi.org/10.1186/s13705-021-00320-6.
[71] B. Strzelecki, Youth engagement in the multilateral energy space in 2019–2021, Development 65 (1) (2022) 48–53, https://doi.org/10.1057/s41301-022-00328-1.
[72] L. Dunlop, L. Atkinson, M. Turkenburg-van Diepen, "It's our future." Youth and fracking justice in England, Local Environ. 26 (1) (2021) 110–130, https://doi.org/10.1080/13549839.2020.186 7837.
[73] N. Mouter, R.M. Shortall, S.L. Spruit, A.V. Itten, Including young people, cutting time and producing useful outcomes: participatory value evaluation as a new practice of public participation in the Dutch energy transition, Energy Res. Soc. Sci. 75 (2021), 101965, https://doi.org/10.1016/j.erss.2021.101965.
[74] J. Radtke, D. Ohlhorst, Community energy in Germany – bowling alone in elite clubs? Util. Policy 72 (2021), 101269 https://doi.org/10.1016/j.jup.2021.101269.
[75] S.D. Kirby, A.G. Guin, A.G. Chilcote, Creating the next generation of residential energy stewards using the energy transformation curriculum, Housing Soc. 42 (3) (2015) 250–258, https://doi.org/10.1080/08882746.2015.1121687.
[76] E. Eaton, N.A. Day, Petro-pedagogy: fossil fuel interests and the obstruction of climate justice in public education, Environ. Educ. Res. 26 (4) (2020) 457–473, https://doi.org/10.1080/13504622.2019.1650164.
[77] H.S.K. Nathan, S.D. Das, A.P. P, Rural microgrids – 'Tragedy of commons' or 'community collective action.', J. Environ. Policy Plan. 24 (4) (2022) 391–406, https://doi.org/10.1080/1523908X.2021.2022466.
[78] R. Shirley, C.J. Lee, H.N. Njoroge, S. Odera, P.K. Mwanzia, I. Malo, Y. Dipo-Salami, Powering jobs: the employment footprint of decentralized renewable energy technologies in sub-Saharan Africa, J. Sustain. Res. 2 (1) (2019), https://doi.org/10.20900/jsr20200001.
[79] C. Jana, Sustainable domestic lighting options for poor people—an empirical study, Environ. Dev. Sustain. 18 (6) (2016) 1559–1573, https://doi.org/10.1007/s10668-015-9702-6.
[80] R. Benedikter, W.M. Tsedze, K. Unterkircher, Africa, Go Green! A new initiative for the continent's youth to become leaders in the global environment needs to combine activism with knowledge, research and policy, New Glob. Stud. 14 (1) (2020) 83–93, https://doi.org/10.1515/ngs-2019-0026.
[81] M.D. Gergan, A. Curley, Indigenous youth and decolonial futures: energy and environmen- talism among the Diné in the Navajo nation and the Lepchas of Sikkim, India, Antipode (2021), 12763, https://doi.org/10.1111/anti.12763.
[82] H. Faheem, Evans Wadongo: bridging the electricity divide in Kenya, Emerald Emerg. Mark. Case Stud. 5 (2) (2015) 1–12, https://doi.org/10.1108/EEMCS-05-2014-0140.
[83] A.N. Aziz, F. Tegegne, J. Koff, R. Wiemers, Biofuel training workshops for diverse stakeholders ranging from pre-college students to educators, Nat. Sci. Educ. 46 (1) (2017) 1–5, https://doi.org/10.4195/nse2016.11.0033.
[84] J. Laquatra, M. Pierce, N. Helmholdt, The consumer education program for residential energy efficiency, J. Ext. 47 (6) (2009) 1–12.
[85] J.T. Bertrand, P. Goldman, N. Zhivan, Y. Agyeman, E. Barber, Evaluation of the "lose your excuse" public service advertising campaign for tweens to save energy, Eval. Rev. 35 (5) (2011) 455–489, https://doi.org/10.1177/0193841X11428489.
[86] A. Zyadin, A. Puhakka, P. Halder, P. Ahponen, P. Pelkonen, The relative importance of home, school, and traditional mass media sources in elevating

youth energy awareness, Appl. Energy 114 (2014) 409–416, https://doi.org/10.1016/j.apenergy.2013.09.072.
[87] R.X. Valdez, M.N. Peterson, K.T. Stevenson, How communication with teachers, family and friends contributes to predicting climate change behaviour among adolescents, Environ. Conserv. 45 (2) (2018) 183–191, https://doi.org/10.1017/S0376892917000443.
[88] B. Frączek, A. Gagat-Matuła, K. Plutecka, Parental attitudes and awareness of sustainable energy consumption in the opinion of young people with Autistic-Spectrum Disorder (ASD): implications for the family budget, Energies 15 (5) (2022) 1730, https://doi.org/10.3390/en15051730.
[89] M. Cornelius, K.C. Armel, K. Hoffman, L. Allen, S.W. Bryson, M. Desai, T. N. Robinson, Increasing energy- and greenhouse gas-saving behaviors among adolescents: a school-based cluster-randomized controlled trial, Energ. Effic. 7 (2) (2014) 217–242, https://doi.org/10.1007/s1 2053-013-9219-5.
[90] T. Ouariachi, M.D. Olvera-Lobo, J. Gutiérrez-Pérez, E. Maibach, A framework for climate change engagement through video games, Environ. Educ. Res. 25 (5) (2019) 701–716, https://doi.org/10.1080/13504622.2018.1545156.
[91] J.J.B.R. Aruta, Science literacy promotes energy conservation behaviors in Filipino youth via climate change knowledge efficacy: evidence from PISA 2018, Aust. J. Environ. Educ. (2022) 1–12, https://doi.org/10.1017/aee.2022.10.
[92] S. Levine Rose, A. Calabrese Barton, Should Great Lakes city build a new power plant? How youth navigate socio-scientific issues, J. Res. Sci. Teach. 49 (5) (2012) 541–567, https://doi.org/10.1002/tea.21017.
[93] R. Brain, S. Upton, B. Tingey, Extension sustainability camp: design, implementation, and evaluation, J. Ext. 53 (1) (2015) 1–18, https://doi.org/10.34068/joe.53.01.06.
[94] A.J. Wensing, E.J. Wensing, M. Virgo, Towards a core curriculum for civic engagement on appropriate technology: characterizing, optimizing and mobilizing youth community service learning, Afr. J. Sci. Technol. Innov. Dev. 10 (7) (2018) 867–877, https://doi.org/10.1080/20421338.2018.1439279.
[95] T. Mori, T. Tasaki, Factors influencing pro-environmental collaborative collective behaviors toward sustainability transition – a case of renewable energy, Environ. Educ. Res. 25 (4) (2019) 566–584, https://doi.org/10.1080/13504622.2018.1545155.
[96] P. Halder, J. Pietarinen, S. Havu-Nuutinen, P. Pelkonen, Young citizens' knowledge and perceptions of bioenergy and future policy implications, Energy Policy 38 (6) (2010) 3058–3066, https://doi.org/10.1016/j.enpol.2010.01.046.
[97] P. Halder, J. Pietarinen, S. Havu-Nuutinen, P. Pelkonen, C.-Y. Chang, P. Prokop, M. Usak, Knowledge, perceptions, and attitudes as determinants of youths' intentions to use bioenergy—a cross-national perspective, Int. J. Green Energy 10 (8) (2013) 797–813, https://doi.org/10.1080/15435075.2012.706244.
[98] A. Assali, T. Khatib, A. Najjar, Renewable energy awareness among future generation of Palestine, Renew. Energy 136 (2019) 254–263, https://doi.org/10.1016/j.renene.2019.01.007.
[99] M. Woźniak, K. Kud, A. Badora, L. Woźniak, Electricity production and consumption perspectives in the opinion of the youth of south-eastern Poland, Energies 15 (13) (2022) 4776, https://doi.org/10.3390/en15134776.
[100] P. Halder, Perceptions of energy production from forest biomass among school students in Finland: directions for the future bioenergy policies, Renew. Energy 68 (2014) 372–377, https://doi.org/10.1016/j.renene.2014.01.051.
[101] S.A. Malik, A.R. Ayop, Solar energy technology: knowledge, awareness, and acceptance of B40 households in one district of Malaysia towards government initiatives, Technol. Soc. 63 (2020), 101416, https://doi.org/10.1016/j.techsoc.2020.101416.
[102] N.P. Simpson, C.J. Rabenold, M. Sowman, C.D. Shearing, Adoption rationales and effects of off-grid renewable energy access for African youth: a case study from Tanzania, Renew. Sust. Energ. Rev. 141 (2021), 110793, https://doi.org/10.1016/j.rser.2021.110793.
[103] X. Liu, Public perception and attitude of the young generation towards nuclear energy and its safety in South Korea, Atoms Peace 3 (4) (2013) 261, https://doi.org/10.1504/AFP.2013.058570.
[104] B.O. Menyeh, Financing electricity access in Africa: a choice experiment study of household investor preferences for renewable energy investments in Ghana, Renew. Sust. Energ. Rev. 146 (2021) (2021), 111132, https://doi.org/10.1016/j.rser.2021.111132.
[105] K. Rogatka, J. Chodkowska-Miszczuk, J. Biegańska, E. Grzelak-Kostulska, S. Środa-Murawska, Perception of the cultural landscape related to wind parks – generation Y perspective, Mitt. Osterr. Geogr. Ges. 159 (2017) 131–149, https://doi.org/10.23781/moegg159-131.
[106] M. Faulques, J. Bonnet, S. Bourdin, M. Juge, J. Pigeon, C. Richard, Generational effect and territorial distributive justice, the two main drivers for willingness to pay for renewable energies, Energy Policy 168 (2022), 113094, https://doi.org/10.1016/j.enpol.2022.113094.
[107] M. Witkowska-Dabrowska, N. Świdyńska, A. Napiórkowska-Baryła, Attitudes of communities in rural areas towards the development of wind energy, Energies 14 (23) (2021) 8052, https://doi.org/10.3390/en14238052.
[108] A. Alexandru, O. Bica, E. Tudora, C.S. Alecu, C.A. Alexandru, I. Covalcic, Enhancing the quality of learning by using an innovative approach for teaching energy in secondary schools, Int. J. Soc. Behav. Educ. Econ. Bus. Ind. Eng. 4 (9) (2010) 1970–1974.
[109] D. Banciu, A. Alexandru, Greening dimension of learning in secondary schools, J. Green Eng. 1 (2) (2011) 111–127.
[110] M. MacKay, B. Parlee, C. Karsgaard, Youth engagement in climate change action: case study on Indigenous youth at COP24, Sustainability 12 (16) (2020), https://doi.org/10.3390/su12166299.
[111] H. Wallis, C. Klöckner, The transmission of energy-saving behaviours in the family: a multilevel approach to the assessment of aggregated and single energy-saving actions of parents and adolescents, Environ. Behav. 52 (3) (2020) 275–304, https://doi.org/10.1177/001391651880 2342.
[112] J. Kim, M. Jang, D. Shin, Examining the role of population age structure upon residential electricity demand: a case from Korea, Sustainability 11 (14) (2019) 3914, https://doi.org/10.3390/su11143914.
[113] N.S.M. Salleh, Preliminary study on youth lifestyle with regards to energy saving, Int. J. Eng. Technol. 7 (4.35) (2018) 468, https://doi.org/10.14419/ijet.v7i4.35.22864.
[114] L. Schmidt, A. Horta, A. Correia, S. Fonseca, Generational gaps and paradoxes regarding electricity consumption and saving, Nat. Cult. 9 (2) (2014) 183–203.
[115] J. Olson, Youth and climate change: an advocate's argument for holding the US government's feet to the fire, Bull. At. Sci. 72 (2) (2016) 79–84, https://doi.org/10.1080/00963402.2016.1145903.
[116] J.A. Alonso, Globalisation, civil society, and the multilateral system, Dev. Pract. 10 (3–4) (2000) 348–360, https://doi.org/10.1080/09614520050116505.
[117] A. Friberg, On the need for (con) temporary utopias: temporal reflections on the climate rhetoric of environmental youth movements, Time Soc. 31 (1) (2022) 48–68, https://doi.org/10.1177/0961463X21998845.
[118] J. Zilles, S. Marg, Protest and polarisation in the context of energy transition and climate policy in Germany: mindsets and collective identities, German Polit. (2022) 1–22, https://doi.org/10.1080/09644008.2022.2059469.
[119] N.O. Bonsu, J. TyreeHageman, J. Kele, Beyond Agenda 2030: future-oriented mechanisms in localising the sustainable development goals (SDGs), Sustainability 12 (23) (2020) 9797, https://doi.org/10.3390/su12239797.
[120] G. Seyfang, J. Park, A. Smith, A thousand flowers blooming? An examination of community energy in the UK, Energy Policy 61 (2013) 977–989, https://doi.org/10.1016/j.enpol.2013.06.030.
[121] M. De Witt, H. Stefánsson, A. Valfells, Energy security in the Arctic: policies and technologies for integration of renewable energy, Arctic Yearbook (2019) (https://arcticyearbook.com/arctic-yearbook/2019/2019-briefing-notes/329-energy-security-in-the-arctic-policies-and-technologies-for-integration-of-renewable-energy).
[122] K. Furlong, EmPower the Yukon. Using community renewable energy in the transition to energy resilience: a social enterprise approach, Northern Rev. 49 (2020), https://doi.org/10.22584/nr49.2020.016.
[123] Y. Hossain, P.A. Loring, T. Marsik, Defining energy security in the rural North—historical and contemporary perspectives from Alaska, Energy Res. Soc. Sci. 16 (2016) 89–97, https://doi.org/10.1016/j.erss.2016.03.01.
[124] D. Cambou, G. Poelzer, Enhancing energy justice in the Arctic: an appraisal of the participation of Arctic indigenous peoples in the transition to renewable energy, in: Renewable Economies in the Arctic, Routledge, 2021, pp. 184–202.
[125] I. Niankara, Youth interests in the biosphere and sensitivity to nuclear power technology in the UAE: with discussions on open innovation and technological convergence in energy and water sectors, J. Open Innov. Technol. Mark. Complex. 6 (4) (2020) 180, https://doi.org/10.3390/joitmc6040180.
[126] G. Holdmann, D. Pride, G. Poelzer, B.F. Noble, C. Walker, Critical pathways to renewable energy transitions in remote Alaska communities: a comparative analysis, Energy Res. Soc. Sci. 91 (2022), https://doi.org/10.1016/j.erss.2022.102712.
[127] R. Nelson, Beyond dependency: economic development, capacity building, and generational sustainability for Indigenous people in Canada, SAGE Open 9 (3) (2019), https://doi.org/10.1177/2158244019879137.
[128] J. Billimoria, Why young people are key to achieving the SDGs, Retrieved from World Economic Forum website, https://www.weforum.org/agenda/2016/09/why-young-people-are-key-to-achieving-the-sdgs/, 2016.
[129] S. McCarthy, T.L. Morrison, Feds must incorporate Indigenous youth in clean energy push - iPolitics, Retrieved December 11, 2021, from, https://ipolitics.ca/2020/07/14/feds-must-incorporate-indigenous-youth-in-clean-energy-push/, 2020.
[130] R. Seydlitz, S. Laska, D. Spain, E.W. Triche, K.L. bishop., Development and social problems: the impact of the offshore oil industry on suicide and homicide rates, Rural. Sociol. 58 (1) (1993) 93–110, https://doi.org/10.1111/j.1549-0831.1993.tb00484.x.
[131] C.L. Seyfrit, L.C. Hamilton, Who will leave? Oil, migration, and Scottish island youth, Soc. Nat. Resour. 5 (3) (1992) 263–276, https://doi.org/10.1080/08941929209380791.
[132] A. Twum-Antwi, P. Jefferies, L. Theron, et al., Young people's perceptions of identities in a rural oil and gas town experiencing boom-bust economic cycles, J. Appl. Youth Stud 3 (2020) 275–292, https://doi.org/10.1007/s43151-020-00020-6.
[133] A.S. Mohammed, E. Graham, S.K. Dary, Rising expectations and dying hopes: local perceptions of oil and gas extraction in Ghana, Energy Res. Soc. Sci. 88 (2022), https://doi.org/10.1016/j.erss.2022.102529.
[134] M.T. Alozie, The left behind: oil, youth and symbolic violence in the Niger delta, J. Youth Stud. 25 (9) (2022) 1165–1181, https://doi.org/10.1080/13676261.2021.1945561.
[135] M.A. Powell, N. Taylor, A.B. Smith, Constructions of rural childhood: challenging dominant perspectives, Child. Geogr. 11 (1) (2013) 117–131, https://doi.org/10.1080/14733285.2013.743285.
[136] E.R. Sensenig, R. El Helou, Gender Mainstreaming Oil and Natural Gas in Lebanon: Analysing the Current, Developing and Potential Impacts of the Petroleum Sector on Women, Men, Youth and Gender Relations From a Multi-stakeholder Group Perspective, Notre Dame University, Louaize. Faculty of Law & Political Science, 2021 (http://ir.ndu.edu.lb/123456789/1428).

[137] T. Skillington, Climate Change and Intergenerational Justice, Routledge, NY, 2019.
[138] J.H. Armstrong, People and power: expanding the role and scale of public engagement in energy transitions, Energy Res. Soc. Sci. 78 (2021), https://doi.org/10.1016/j.erss.2021.102136.
[139] E. Galende-Sanchez, A.H. Sorman, From consultation toward co-production in science and policy: a critical systematic review of participatory climate and energy initiatives, Energy Res. Soc. Sci. 73 (2021), https://doi.org/10.1016/j.erss.2020.101907.
[140] B.K. Sovacool, D.J. Hess, S. Amir, F.W. Geels, R. Hirsh, L. Rodriguez Medina, C. Miller, C. Alvial Palavicino, R. Phadke, M. Ryghaug, J. Schot, A. Silvast, J. Stephens, A. Stirling, B. Turnheim, E. van der Vleuten, H. van Lente, S. Yearley, Sociotechnical agendas: Reviewing future directions for energy and climate research, Energy Res. Soc. Sci. 70 (2020), 101617.
[141] O. Benchikh, Global renewable energy education and training program (GREET Programme), Desalination 141 (2) (2001) 209–221, https://doi.org/10.1016/S0011-9164(01)00406-4.
[142] M. Alam, A. Rahman, M. Eusuf, Diffusion potential of renewable energy technology for sustainable development: Bangladeshi experience, Energy Sustain. Dev. 7 (2) (2003) 88–96, https://doi.org/10.1016/S0973-0826(08)60358-0.
[143] A. Calabrese Barton, D. Birmingham, T. Sato, E. Tan, S. Calabrese Barton, Youth as community science experts in green energy technologies, Aftersch. Matters (2013) 25–32.
[144] J. Baka, S. Vaishnava, The evolving borderland of energy geographies, Geogr. Compass 14 (7) (2020), https://doi.org/10.1111/gec3.12493.
[145] A. Doyon, J. Boron, S. Williams, Unsettling transitions: representing indigenous peoples and knowledge in transitions research, Energy Res. Soc. Sci. 81 (2021), 102255, https://doi.org/10.1016/j.erss.2021.102255.
[146] C.E. Hoicka, K. Savic, A. Campney, Reconciliation through renewable energy? A survey of Indigenous communities, involvement, and peoples in Canada, Energy Res. Soc. Sci. 74 (2021), 101897, https://doi.org/10.1016/j.erss.2020.101897.
[147] R. Rakshit, C. Shahi, M.A. Smith, A. Cornwell, Energy transition complexities in rural and remote Indigenous communities: a case study of Poplar Hill First Nation in northern Ontario, Local Environ. 24 (9) (2019) 809–824, https://doi.org/10.1080/13549839.2019.1648400.
[148] N. Mercer, P. Parker, A. Hudson, D. Martin, Off-grid energy sustainability in Nunatukavut, Labrador: centering Inuit voices on heat insecurity in diesel-powered communities, Energy Res. Soc. Sci. 62 (2020), 101382, https://doi.org/10.1016/j.erss.2019.101382.
[149] Q.V. Khuc, M. Tran, T. Nguyen, N.A. Thinh, T. Dang, T.D. Tuyen, P. Pham, L. Q. Dat, Improving energy literacy to facilitate energy transition and nurture environmental culture in Vietnam, Urban Sci. 7 (1) (2023), https://doi.org/10.3390/urbansci7010013.
[150] B. Kaczmarczyk, K. Lis, Perception of the transition to a zero-emission economy in the opinion of Polish and foreign students: a comparative analysis, Energies 16 (18) (2023) 6486, https://doi.org/10.3390/en16186486.
[151] S. Holzer, A. Dubois, J. Cousse, G. Xexakis, E. Trutnevyte, Swiss electricity supply scenarios: perspectives from the young generation, Energy Clim. Chang. 4 (2023), 100109, https://doi.org/10.1016/j.egycc.2023.100109.